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defined as

$$\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$
Where $\hat{y}_i$ is the prediction for i^{th} data point.

Enter your solution here

Question 8
Find the weights using the gradient descent. Use a constant learning rate of $\eta = 10^{-10}$. Initialize the weight vector as zero vector and update the weights for 100 iterations. Enter the sum of all the weights.

Enter your solution here

Question 9
Find the loss for the training data points if the model is learnt using the gradient descent as in question 8. Consider the loss to be defined as

$$\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$
Where $\hat{y}_i$ is the prediction for i^{th} data point.

Enter your solution here

Question 10
Find the loss for the test data points if the model is learnt using the gradient descent as in question 8. Consider the loss to be defined as

$$\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$
Where $\hat{y}_i$ is the prediction for i^{th} data point.

Enter your solution here

Question 11
Find the weights using the stochastic gradient descent. Use a constant learning rate of $\eta = 10^{-8}$. Initialize the weight vector as zero vector and update the weights for 1000 iterations. . Take the batch size of $\lceil \frac{\text{number of samples}}{5} \rceil$. For sampling the batch examples in i^{th} iteration, set seed at i . The final weight is the last updated weight. Do not take the average of weights updated in all the iterations. Enter the sum of all the weights.

Enter your solution here

Question 12
Find the loss for the training data points if the model is learnt using the stochastic gradient descent as in question 11. Consider the loss to be defined as

$$\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$
Where $\hat{y}_i$ is the prediction for i^{th} data point.

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